```
# Install required libraries
!pip install transformers torch

# Import libraries
from transformers import pipeline
import pandas as pd
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->tr
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggi
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

# Create a custom summarizer class to mimic the pipeline behavior
class CustomSummarizer:
    def __init__(self, model, tokenizer):
        self.model = model
        self.tokenizer = tokenizer

    def __call__(self, text, max_length, min_length, do_sample):
        inputs = self.tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)
        # Generate Summary
        summary_ids = self.model.generate(
            inputs["input_ids"],
            num_beams=4,
            max_length=max_length,
            min_length=min_length,
            early_stopping=True
        )
        summary_text = self.tokenizer.decode(summary_ids[0], skip_special_tokens=True)
        return [{'generated_text': summary_text}]

summarizer = CustomSummarizer(model, tokenizer)
print("Model Loaded Successfully!")
```

```
Loading weights: 100%                    511/511 [00:00<00:00, 745.23it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm.weight]
Model Loaded Successfully!
```

Start coding or generate with AI.

Start coding or generate with AI.

```
texts = [
    """Artificial Intelligence is transforming industries by enabling machines
    to learn from data and perform tasks that traditionally required human intelligence.
    Applications include healthcare diagnosis, recommendation systems, and automation.
    However, ethical challenges such as bias and privacy remain concerns.""",

    """Climate change is one of the biggest global challenges. Rising temperatures,
    melting glaciers, and extreme weather events affect ecosystems and human life.
    Governments are promoting renewable energy and sustainability initiatives.""",

    """Python is a popular programming language known for readability and simplicity.
    It is widely used in web development, data science, machine learning, and automation.
    Its extensive libraries make development efficient.""",

    """Once upon a time, a young explorer traveled across oceans searching for
    hidden civilizations. Along the journey, challenges shaped his courage
    and wisdom, ultimately leading him to discover the value of friendship.""",

    """Hey, did you finish the assignment? Not yet, I was working on the coding part.
    The professor said we must also write analysis and compare outputs before submission."""
]
```

```
results = []

for text in texts:
    summary = summarizer(text, max_length=60, min_length=20, do_sample=False)
    results.append(summary[0]['generated_text'])

# Display results
for i in range(len(texts)):
    print("\nOriginal Text:\n", texts[i])
    print("\nSummary:\n", results[i])
    print("-"*80)
```

```
Original Text:
 Artificial Intelligence is transforming industries by enabling machines
    to learn from data and perform tasks that traditionally required human intelligence.
    Applications include healthcare diagnosis, recommendation systems, and automation.
    However, ethical challenges such as bias and privacy remain concerns.

Summary:
 Artificial Intelligence is transforming industries by enabling machines
    to learn from data and perform tasks that traditionally required human intelligence.
    Applications include healthcare diagnosis, recommendation systems, and automation.
    However, ethical challenges such as bias and privacy remain concerns. "
--------------------------------------------------------------------------------

Original Text:
 Climate change is one of the biggest global challenges. Rising temperatures,
    melting glaciers, and extreme weather events affect ecosystems and human life.
    Governments are promoting renewable energy and sustainability initiatives.

Summary:
 Climate change is one of the biggest global challenges. Rising temperatures,
    melting glaciers, and extreme weather events affect ecosystems and human life.
    Governments are promoting renewable energy and sustainability initiatives.After only only onlyo one only onlyTe onlyo onlyo
--------------------------------------------------------------------------------

Original Text:
 Python is a popular programming language known for readability and simplicity.
    It is widely used in web development, data science, machine learning, and automation.
    Its extensive libraries make development efficient.

Summary:
 Python is a popular programming language known for readability and simplicity.
    It is widely used in web development, data science, machine learning, and automation.
    Its extensive libraries make development efficient. un un unun still un un only un un had un un
--------------------------------------------------------------------------------

Original Text:
 Once upon a time, a young explorer traveled across oceans searching for
    hidden civilizations. Along the journey, challenges shaped his courage
    and wisdom, ultimately leading him to discover the value of friendship.

Summary:
 Once upon a time, a young explorer traveled across oceans searching for
    hidden civilizations. Along the journey, challenges shaped his courage
```

```
    and wisdom, ultimately leading him to discover the value of friendship. still only only only had only onlyShe only only Only
--------------------------------------------------------------------------------

Original Text:
 Hey, did you finish the assignment? Not yet, I was working on the coding part.
    The professor said we must also write analysis and compare outputs before submission.

Summary:
 Hey, did you finish the assignment? Not yet, I was working on the coding part.
    The professor said we must also write analysis and compare outputs before submission. one had only only had had had one had
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```

```
text_sample = texts[0]

short_summary = summarizer(text_sample, max_length=30, min_length=10, do_sample=False)
medium_summary = summarizer(text_sample, max_length=60, min_length=20, do_sample=False)
long_summary = summarizer(text_sample, max_length=100, min_length=40, do_sample=False)

print("Short Summary:\n", short_summary[0]['generated_text'])
print("\nMedium Summary:\n", medium_summary[0]['generated_text'])
print("\nLong Summary:\n", long_summary[0]['generated_text'])
```

```
Short Summary:
 Artificial Intelligence is transforming industries by enabling machines to learn from data. Applications include healthcare dia

Medium Summary:
 Artificial Intelligence is transforming industries by enabling machines to learn from data. Applications include healthcare dia

Long Summary:
 Artificial Intelligence is transforming industries by enabling machines  to learn from data. Applications include healthcare di
```

```
long_text = " ".join(texts) * 3   # create long 400+ word text

long_summary = summarizer(long_text[:1000], max_length=120, min_length=50, do_sample=False)

print(long_summary[0]['generated_text'])
```

```
Artificial Intelligence is transforming industries by enabling machines to learn from data and perform tasks that traditionally
```